

Notes on coupling RAPTOR to equilibrium code

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1 Outline of coupling between RAPTOR and GS reconstruction code

- LIUQE is a Grad-Shafranov equilibrium reconstruction code that computes $\psi(R, Z)$ given measurements of currents, fields, voltages, etc. The reconstruction of internal details of the pressure and current density profiles can not be resolved due to lack of internal measurements.
- RAPTOR is a transport code that evolves ψ and p on a $\hat{\rho} = \rho_{tor, N} = \sqrt{\Phi/\Phi_b}$ grid, with given geometric information about the flux surfaces.

This document outlines the coupling between the two codes, to have LIUQE use RAPTOR's detailed solution for the current density and pressure profile, while RAPTOR uses LIUQE's flux surface geometry information.

2 Computing geometric profiles from Grad-Shafranov equilibria

2.1 Quantities required by RAPTOR

For solving the transport equations, RAPTOR requires

- $F = RB_\phi$,
- $\Phi = \int_S B_\phi dS_\phi$
- $g_1 = \langle (\nabla V)^2 \rangle = \left(\frac{\partial V}{\partial \psi} \right)^2 \langle |\nabla \psi|^2 \rangle$
- $\frac{g_2 g_3}{\rho} = \frac{\left\langle \frac{(\nabla V)^2}{R^2} \right\rangle \left\langle \frac{1}{R^2} \right\rangle}{\rho} = \frac{1}{\rho} \left(\frac{\partial V}{\partial \psi} \right)^2 \left\langle \frac{|\nabla \psi|^2}{R^2} \right\rangle \left\langle \frac{1}{R^2} \right\rangle$
- $\frac{\partial V}{\partial \hat{\rho}} = \frac{4\pi\Phi_b\hat{\rho}}{F\langle 1/R^2 \rangle}$,

For later calculating TT' from the RAPTOR output, we also need

$$\frac{\partial \hat{\rho}}{\partial V} g_2 = \frac{\partial \hat{\rho}}{\partial V} \left\langle \frac{(\nabla V)^2}{R^2} \right\rangle = \left\langle \frac{\nabla \rho \nabla \psi}{R^2} \right\rangle \frac{\partial V}{\partial \psi} \quad (1)$$

Recall that $\langle Q \rangle$ is defined as

$$\langle Q \rangle = \frac{\oint Q \frac{dl_p}{B_p}}{\oint \frac{dl_p}{B_p}} \quad (2)$$

2.2 Contour integrals on the 2D equilibrium

To compute the desired quantities we need compute integrals of the form

$$C_i = \oint |\nabla\psi|^\mu R^\nu dl \quad (3)$$

We use the definitions from CHEASE [1], using COCOS [2] for the sign convention, and also refer to the definitions from the LIUQE paper [3].

$$C_0 = \oint \frac{1}{R} \frac{dl_p}{B_p} = (2\pi)^{e_{Bp}} \oint \frac{dl_p}{|\nabla\psi|} = Q_{-1,0} \quad (4)$$

$$C_1 = \oint \frac{dl_p}{B_p} = (2\pi)^{e_{Bp}} \oint \frac{R dl_p}{|\nabla\psi|} = Q_{-1,1} \quad (5)$$

$$C_2 = \oint \frac{1}{R^2} \frac{dl_p}{B_p} = (2\pi)^{e_{Bp}} \oint \frac{dl_p}{R|\nabla\psi|} = Q_{-1,-1} \quad (6)$$

$$C_3 = \oint B_p^2 \frac{dl_p}{B_p} = \oint B_p dl_p = \frac{1}{(2\pi)^{e_{Bp}}} \oint \frac{|\nabla\psi|}{R} dl_p = Q_{1,-1} \quad (7)$$

$$C_4 = \oint R^2 B_p^2 \frac{dl_p}{B_p} = \oint R^2 B_p dl_p = \frac{1}{(2\pi)^{e_{Bp}}} \oint R |\nabla\psi| dl_p = Q_{1,1} \quad (8)$$

2.3 Limit cases

For cases outside the axis or last closed flux surface, these can be computed by numerical integration around the contour. The accuracy is determined by the spatial grid resolution and numerical contour integration method. For the axis and LCFS, problems may occur in particular for cases with $\mu < 0$.

- At the axis, both dl and $|\nabla\psi|^\mu$ go to infinity, and the result is ill-defined.
- At the LCFS, $|\nabla\psi|^\mu$ goes to infinity and the integral goes to infinity.

We avoid these problems by directly calculating the values of the flux-surface averages, which are well-defined in all cases.

2.3.1 Values close to the magnetic axis

- Flux surface averages of quantities involving R go to their value at the axis in the limit. For example $\lim_{\psi \rightarrow \psi_a} \langle 1/R^2 \rangle = 1/R_a^2$.
- Flux surface averages of quantities involving $\nabla\psi$ go to zero since the flux has an extremum at the axis.
- Similarly $\lim_{\psi \rightarrow \psi_a} \left\langle \frac{\nabla\rho \nabla\psi}{R^2} \right\rangle \frac{\partial V}{\partial \psi} = 0$ since $\nabla\psi = 0$ and all other quantities in the expression stay finite.

- Values of $V\psi$ at the axis can be computed by approximating the contour lines as elliptic, corresponding to a quadratic Taylor expansion around the axis [3]:

$$\lim_{\psi \rightarrow \psi_a} \frac{\partial V}{\partial \psi} = \text{sign}(\psi_b - \psi_a) \frac{4\pi^2 R_a}{\sqrt{\det(H)}} \quad (9)$$

where $H(\psi_a)$ is the Hessian of the quadratic approximation of the flux $\psi(R, Z)$ near the axis.

2.3.2 Values at the LCFS for diverted plasmas

At the LCFS, the denominator of $\frac{d\ell}{B_p}$ goes to zero close to the $\times$ -point. Effectively this means that the value of the flux-surface-averaged quantity (2) is weighed ‘infinitely more’ at the $\times$ -point and the value is simply the value at the $\times$ -point. For example:

$$\lim_{\psi \rightarrow \psi_b} \langle 1/R \rangle = 1/R_\times \quad (10)$$

$$\lim_{\psi \rightarrow \psi_b} \langle (\nabla \psi)^2 \rangle = 0 \quad (11)$$

Special care needs to be taken to compute Φ_b , the total toroidal flux enclosed by the plasma. Since q diverges at the LCFS for diverted plasmas, numerical integration can be tricky. It is better to compute this quantity directly from the 2D equilibrium using

$$\Phi = \int_S B_\psi dS_\phi = F(\psi) \int_S \frac{dS_\phi}{R} \quad (12)$$

The value of the integral $\int_S \frac{dS_\phi}{R}$ can be computed for the LCFS from the 2D $\psi(R, Z)$ solution.

2.3.3 Summary of output quantities required from equilibrium code

Scalar quantities:

- ψ_a
- ψ_b
- $\int_S \frac{dS_\phi}{R}$

Profile quantities (on a fixed grid of ψ_N):

Quantity	On axis	close to axis	Flux surfaces without $\times$ -point	Separatrix with $\times$ -point
$F(\psi)$	$F(\psi_a)$	-	$F(\psi)$	$F(\psi_b)$
$\langle 1/R \rangle$	$1/R_a$	$1/R_a$	C_0/C_1	$1/R_\times$
$\langle 1/R^2 \rangle$	$1/R_a^2$	?	C_2/C_1	$1/R_\times^2$
$\langle \nabla\psi ^2/R^2 \rangle$	0	?	C_3/C_1	0
$\langle \nabla\psi ^2 \rangle$	0	?	C_4/C_1	0
$\frac{\partial\psi}{\partial V}$	$\text{sign}(\psi_b - \psi_a) \frac{\sqrt{\det(H)}}{4\pi^2 R_a}$	?	$1/C_1$	0

Table 1: Summary of required output quantities from equilibrium code. For limited plasmas the LCFS value can be computed as for any flux surface, for diverted plasmas one should use the special value in the rightmost column.

2.4 Computing RAPTOR quantities from provided equilibrium profiles

- $\Phi = \frac{T(\psi)}{2\pi} \frac{\partial V}{\partial \psi} \int g_3 d\psi$ for $\psi < \psi_b$, $\Phi(\psi_b) = F(\psi_b) \int_S \frac{dS_\phi}{R}$
- ...

3 Feeding the RAPTOR profiles to the GS equilibrium

3.1 Computing p' and FF' from RAPTOR

We use the notes by A. Merle for the ASTRA sawtooth module.

Starting from the GS equation in the form:

$$\nabla \cdot \left(\frac{\nabla \psi}{R^2} \right) = -4\pi^2 \left(\mu_0 p' + \frac{1}{R^2} FF' \right) \quad (13)$$

Taking the flux surface average and using $\langle \nabla \cdot F \rangle = \frac{\partial}{\partial V} \langle F \cdot \nabla V \rangle$:

$$\left\langle \nabla \cdot \left(\frac{\nabla \psi}{R^2} \right) \right\rangle = -4\pi^2 \left(\mu_0 p' + \left\langle \frac{1}{R^2} \right\rangle FF' \right) \quad (14)$$

$$\frac{\partial}{\partial V} \left\langle \frac{\nabla \psi}{R^2} \cdot \nabla V \right\rangle = -4\pi^2 \left(\mu_0 p' + \left\langle \frac{1}{R^2} \right\rangle FF' \right) \quad (15)$$

$$\frac{\partial \hat{\rho}}{\partial V} \frac{\partial}{\partial \hat{\rho}} \left(\frac{\partial \psi}{\partial V} \left\langle \frac{(\nabla V)^2}{R^2} \right\rangle \right) = -4\pi^2 \left(\mu_0 p' + \left\langle \frac{1}{R^2} \right\rangle FF' \right) \quad (16)$$

$$\frac{\partial \hat{\rho}}{\partial V} \frac{\partial}{\partial \hat{\rho}} \left(\frac{\partial \psi}{\partial \hat{\rho}} g_2 \frac{\partial \hat{\rho}}{\partial V} \right) = -4\pi^2 \left(\mu_0 p' + \left\langle \frac{1}{R^2} \right\rangle FF' \right) \quad (17)$$

$$FF' = -\frac{1}{\left\langle \frac{1}{R^2} \right\rangle} \left(\frac{1}{4\pi^2} \frac{\partial \hat{\rho}}{\partial V} \frac{\partial}{\partial \hat{\rho}} \left(\frac{\partial \psi}{\partial \hat{\rho}} g_2 \frac{\partial \hat{\rho}}{\partial V} \right) + \mu_0 p' \right) \quad (18)$$

$$= -\frac{1}{g_3} \left(\frac{1}{4\pi^2} \frac{1}{V'} \left(\frac{\partial \psi}{\partial \hat{\rho}} \left(\frac{1}{V'} \frac{\partial g_2}{\partial \rho} - \frac{V'' g_2}{V'^2} \right) + \frac{g_2}{V'} \frac{\partial^2 \psi}{\partial \rho^2} \right) + \mu_0 p' \right) \quad (19)$$

where the blue quantities are taken from the current RAPTOR time step and the magenta quantities are taken from the previous equilibrium solution.

These quantities are returned on the RAPTOR ρ grid, together with the corresponding ψ values.

3.2 Computing the measurement response due to the RAPTOR current density

FF' and p' are used to calculate a current density $\mathbf{j}(R, Z)$ as right hand side for the GS equation:

$$\mathbf{j}_{\phi, \text{RAPTOR}} = -4\pi \left(p'(\psi(R, Z)) + \frac{1}{R^2} FF'(\psi(R, Z)) \right) \quad (20)$$

The measurement response caused by this current density is computed as

$$m_{RAP} = T_{meas, x} j_{RAP} \quad (21)$$

3.3 Inclusion in LIUQE measurement equation

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A Analytical approximation of flux integrals close to the axis (A. Palha)

Close to the magnetic axis it is possible to construct a quadratic approximation of the magnetic flux based on its Taylor expansion.

A.1 Approximate magnetic flux

The second order terms of the Taylor expansion of the magnetic flux are given by:

$$\psi(r, z) = \psi_A + \frac{1}{2} \begin{bmatrix} r - r_A & z - z_A \end{bmatrix} \mathbf{H}_{(r_A, z_A)} \begin{bmatrix} r - r_A \\ z - z_A \end{bmatrix}. \quad (22)$$

Where (r_A, z_A) are the coordinates of the magnetic axis, $\psi_A = \psi(r_A, z_A)$ and $\mathbf{H}_{(r_A, z_A)}$ is the Hessian matrix evaluated at the magnetic axis:

$$\mathbf{H}_{(r_A, z_A)} := \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} = \begin{bmatrix} \left. \frac{\partial^2 \psi}{\partial r^2} \right|_{(r_A, z_A)} & \left. \frac{\partial^2 \psi}{\partial r \partial z} \right|_{(r_A, z_A)} \\ \left. \frac{\partial^2 \psi}{\partial r \partial z} \right|_{(r_A, z_A)} & \left. \frac{\partial^2 \psi}{\partial z^2} \right|_{(r_A, z_A)} \end{bmatrix}. \quad (23)$$

Note that $h_{12} = h_{21}$.

If we expand equation 22 we obtain:

$$\psi(r, z) = \psi_A + \frac{1}{2} h_{11} (r - r_A)^2 + h_{12} (r - r_A)(z - z_A) + \frac{1}{2} h_{22} (z - z_A)^2. \quad (24)$$

A.2 Approximate contour lines

To compute the parametric expressions of the contour lines of expression 24 we first rewrite this expression in the following form:

$$\Delta\psi(r, z) = \frac{1}{2} h_{11} \Delta r^2 + h_{12} \Delta r \Delta z + \frac{1}{2} h_{22} \Delta z^2. \quad (25)$$

Where $\Delta\psi(r, z) = \psi(r, z) - \psi_A$, $\Delta r = r - r_A$ and $\Delta z = z - z_A$.

An isoline associated to a value of magnetic flux equal to ψ_C corresponds to the points defined implicitly by the following expression:

$$\Delta\psi_C = \frac{1}{2} h_{11} \Delta r^2 + h_{12} \Delta r \Delta z + \frac{1}{2} h_{22} \Delta z^2. \quad (26)$$

Where $\Delta\psi_C = \psi_C - \psi_A$.

We focus on the case where $h_{12}^2 < h_{11} h_{22}$, which corresponds to elliptical isolines. We can transform this equation into the quadratic form of an ellipse:

$$1 = \frac{h_{11}}{2\Delta\psi_C} \Delta r^2 + \frac{h_{12}}{\Delta\psi_C} \Delta r \Delta z + \frac{h_{22}}{2\Delta\psi_C} \Delta z^2. \quad (27)$$

The parametric form of an ellipse (the isolines) is given by the two expressions:

$$\begin{cases} \Delta r(\theta) = a \cos \theta \cos \varphi - b \sin \theta \sin \varphi \\ \Delta z(\theta) = a \cos \theta \sin \varphi + b \sin \theta \cos \varphi \end{cases} \quad (28)$$

Where a is half of the major axis of the ellipse, b is half of the minor axis of the ellipse and φ is the angle between the major axis and the x -axis. Therefore, the computation of the parametric expressions of the isoline resumes to determining a , b and φ .

The major axis a and minor axis b are related to the matrix $\mathbf{H}$ in the following way. If λ_1, λ_2 are the eigenvalues of $\mathbf{H}$ such that $|\lambda_1| < |\lambda_2|$, then:

$$\begin{cases} a = \sqrt{\left| \frac{2\Delta\psi_C}{\lambda_1} \right|} \\ b = \sqrt{\left| \frac{2\Delta\psi_C}{\lambda_2} \right|} \end{cases}. \quad (29)$$

If $\vec{v}_1$ and $\vec{v}_2$ are the eigenvectors associated to the eigenvalues λ_1 and λ_2 , respectively, then the angle φ is the angle $\vec{v}_1$ makes with the x -axis. Therefore we can write: $\vec{v}_1 = \cos \varphi \vec{e}_x + \sin \varphi \vec{e}_2$. The angle φ is then obtained by solving the expression:

$$\begin{bmatrix} h_{11} - \lambda_1 & h_{12} \\ h_{12} & h_{22} - \lambda_1 \end{bmatrix} \begin{bmatrix} \cos \varphi \\ \sin \varphi \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}. \quad (30)$$

Which results in:

$$\begin{cases} \cos \varphi = \frac{\text{sign}(h_{11})|h_{12}|}{\sqrt{h_{12}^2 + (h_{11} - \lambda_1)^2}} \\ \sin \varphi = \frac{-|h_{12}|}{\sqrt{h_{12}^2 + (h_{22} - \lambda_1)^2}} \end{cases}. \quad (31)$$

In the special case $h_{12} = 0$ then:

$$\cos \varphi = \begin{cases} 0, & \text{if } |h_{11}| \geq |h_{22}| \\ 1, & \text{if } |h_{11}| < |h_{22}| \end{cases} \quad \text{and} \quad \sin \varphi = \begin{cases} 1, & \text{if } |h_{11}| \geq |h_{22}| \\ 0, & \text{if } |h_{11}| < |h_{22}| \end{cases}. \quad (32)$$

A.3 Computation of the line integrals

A.3.1 $c_0 = \oint \frac{dl}{|\nabla\psi|}$

The computation of the integral $c_0 = \oint \frac{dl}{|\nabla\psi|}$ is most easily done by noting that this integral is invariant to the orientation of the isolines (ellipses). This is due to the fact that $\nabla\psi$ is a quantity that depends only on a and b . For this reason, we can compute this integral for the simpler case $\varphi = 0$. Explicitly, this integral is given by:

$$c_0 = \oint \frac{dl}{|\nabla\psi|} = \int_0^{2\pi} \frac{d\theta |\dot{\vec{\gamma}}(\theta)|}{|\nabla\psi|}. \quad (33)$$

Where $\vec{\gamma}(\theta)$ is the parameterization of the isoline, given by equation 28 and $\dot{\vec{\gamma}}(\theta) = \frac{d\vec{\gamma}}{d\theta}$. The term $|\nabla\psi|$ is obtained by computing the gradient of equation 25:

$$|\nabla\psi| = \sqrt{2\Delta\psi_C} \sqrt{h_{11} \cos^2(2\pi\theta) + h_{22} \sin^2(2\pi\theta)} \quad (34)$$

Substituting these expressions into equation 33 we obtain:

$$c_0 = \int_0^{2\pi} \frac{d\theta |\dot{\gamma}(\theta)|}{|\nabla\psi|} \quad (35)$$

$$= \int_0^{2\pi} \frac{\sqrt{2\Delta\psi_C} \sqrt{\frac{\cos^2 \theta}{h_{22}} + \frac{\sin^2 \theta}{h_{11}}}}{\sqrt{2\Delta\psi_C} \sqrt{h_{11} \cos^2 \theta + h_{22} \sin^2 \theta}} d\theta \quad (36)$$

$$= \int_0^{2\pi} \sqrt{\frac{h_{11} \cos^2 \theta + h_{22} \sin^2 \theta}{h_{11} h_{22}}} d\theta \quad (37)$$

$$= \int_0^{2\pi} \frac{1}{\sqrt{h_{11} h_{22}}} d\theta \quad (38)$$

This can be generalised to an arbitrary ellipse by rewriting this expression using the eingenvales:

$$c_0 = \frac{2\pi}{\sqrt{\lambda_1 \lambda_2}}. \quad (39)$$

Which, can also be written as a function of the entries of the matrix H:

$$c_0 = \frac{2\pi}{\sqrt{h_{11} h_{22} - h_{12}^2}}. \quad (40)$$

A.3.2 $c_1 = \oint \frac{r(\theta) dl}{|\nabla\psi|}$

The computation of the integral $c_1 = \oint \frac{r(\theta) dl}{|\nabla\psi|}$ is done by directly substituting the terms that appear in it:

$$c_1 = \oint \frac{r(\theta) dl}{|\nabla\psi|} \quad (41)$$

$$= \int_0^{2\pi} \frac{r(\theta) |\dot{\gamma}(\theta)|}{|\nabla\psi|} d\theta. \quad (42)$$

$$(43)$$

From the previous section we know that $\frac{|\dot{\gamma}(\theta)|}{|\nabla\psi|} = \frac{1}{\sqrt{\lambda_1 \lambda_2}}$. Therefore:

$$c_1 = \frac{1}{\sqrt{\lambda_1 \lambda_2}} \int_0^{2\pi} r(\theta) d\theta. \quad (44)$$

If we now note that $r(\theta) = r_A + \Delta r(\theta)$ and use equation 28 we get:

$$c_1 = \frac{1}{\sqrt{\lambda_1 \lambda_2}} \int_0^{2\pi} [r_A + a \cos \theta \cos \varphi - b \sin \theta \sin \varphi] d\theta. \quad (45)$$

The integral of the term including $\sin \theta$ or $\cos \theta$ are zero and we obtain:

$$c_1 = \frac{1}{\sqrt{\lambda_1 \lambda_2}} \int_0^{2\pi} r_A d\theta \quad (46)$$

$$= \frac{2\pi r_A}{\sqrt{\lambda_1 \lambda_2}}. \quad (47)$$

Noting that $\lambda_1 \lambda_2 = \det \mathbf{H} = h_{11} h_{22} - h_{12}^2$, we can rewrite c_1 as:

$$c_1 = \frac{2\pi r_A}{\sqrt{h_{11} h_{22} - h_{12}^2}}. \quad (48)$$

A.3.3 $c_2 = \oint \frac{dl}{r(\theta)|\nabla\psi|}$

The computation of the integral $c_2 = \oint \frac{dl}{r(\theta)|\nabla\psi|}$ is done by directly substituting the terms that appear in it:

$$c_2 = \oint \frac{dl}{r(\theta)|\nabla\psi|} \quad (49)$$

$$= \int_0^{2\pi} \frac{|\dot{\gamma}(\theta)|}{r(\theta)|\nabla\psi|} d\theta. \quad (50)$$

From section A.3.1 we know that $\frac{|\dot{\gamma}(\theta)|}{|\nabla\psi|} = \frac{1}{\sqrt{\lambda_1 \lambda_2}}$. Therefore:

$$c_2 = \frac{1}{\sqrt{\lambda_1 \lambda_2}} \int_0^{2\pi} \frac{1}{r(\theta)} d\theta. \quad (51)$$

We now note again that $r(\theta) = r_A + \Delta r(\theta)$ and use equation 28 to get:

$$c_2 = \frac{1}{\sqrt{\lambda_1 \lambda_2}} \int_0^{2\pi} \frac{1}{r_A + a \cos \theta \cos \varphi - b \sin \theta \sin \varphi} d\theta. \quad (52)$$

This integral can be evaluated yielding:

$$\int_0^{2\pi} \frac{1}{r_A + a \cos \theta \cos \varphi - b \sin \theta \sin \varphi} d\theta = \frac{2\pi}{\sqrt{r_A^2 - a^2 - b^2}}. \quad (53)$$

Therefore c_2 takes the form:

$$c_2 = \frac{2\pi}{\sqrt{\lambda_1 \lambda_2 (r_A^2 - a^2 - b^2)}}. \quad (54)$$

Which can be rewritten as:

$$c_2 = \frac{2\pi}{\sqrt{\lambda_1 \lambda_2 \left(r_A^2 - \left| \frac{\Delta\psi_C}{\lambda_1} \right| - \left| \frac{\Delta\psi_C}{\lambda_2} \right| \right)}}. \quad (55)$$

Once again using $\lambda_1 \lambda_2 = \det \mathbf{H} = h_{11} h_{22} - h_{12}^2$, we can rewrite c_2 as:

$$c_2 = \frac{2\pi}{\sqrt{(h_{11} h_{22} - h_{12}^2) \left(r_A^2 - \left| \frac{\Delta \psi_C}{\lambda_1} \right| - \left| \frac{\Delta \psi_C}{\lambda_2} \right| \right)}}. \quad (56)$$

References

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